

EFFICIENCY, DECOMPOSITION, DATA EXAMPLES Solutions

COMPUTER SCIENCE MENTORS 61A

October 28 – October 31, 2025

1 Efficiency and Orders of Growth

1. Give a tight asymptotic runtime bound for the following functions in $\Theta(\cdot)$ notation, or “Infinite” if the program does not terminate.

(a) `def one(n):`
 `while n > 0:`
 `n = n // 2`

$\Theta(\log n)$

(b) `def two(n):`
 `for i in range(n):`
 `for j in range(i):`
 `print(str(i), str(j))`

$\Theta(n^2)$

(c) `def three(n):`
 `i = 1`
 `while i <= n:`
 `for j in range(i):`
 `print(j)`
 `i *= 2`

$\Theta(n)$

2. What do these runtimes simplify to?

$\Theta(8n^2)$	
$\Theta(10000)$	
$\Theta(100n^2 + 10n^3)$	
$\Theta(3n^2 * 2^n)$	
$\Theta(n^{30} + 3^n)$	
$\Theta(n \log n + \log n + n)$	

$\Theta(8n^2)$	$\Theta(n^2)$
$\Theta(10000)$	$\Theta(1)$
$\Theta(100n^2 + 10n^3)$	$\Theta(n^3)$
$\Theta(3n^2 * 2^n)$	$\Theta(n^2 * 2^n)$
$\Theta(n^{30} + 3^n)$	$\Theta(3^n)$
$\Theta(n \log n + \log n + n)$	$\Theta(n \log n)$

2 Review

1. Fill in the implementation of double_link.

```
def double_link(lnk):
    '''Using mutation, replaces the second in each pair of items
    with the first. The first of each pair stays as is.

    >>> double_link(Link(1, Link(2, Link(3, Link(4)))))
    Link(1, Link(1, Link(3, Link(3))))
    >>> double_link(Link('c', Link('s', Link(6, Link(1, Link('a'))))))
    Link('c', Link('c', Link(6, Link(6, Link('a')))))
    '''

    if lnk is Link.empty or lnk.rest is Link.empty:
        return lnk
    lnk.rest.first = lnk.first
    double_link(lnk.rest.rest)
    return lnk
```

2. Write a function that returns `True` if there exists a path from root to leaf that contains at least `n` instances of `elem` in a tree `t`.

*Hint: recall that the built-in function **any** takes in an iterable and returns `True` if any of the iterable's elements are truthy.*

```
def contains_n(elem, n, t):
    '''
    >>> t1 = Tree(1, [Tree(1, [Tree(2)])])
    >>> contains_n(1, 2, t1)
    True
    >>> contains_n(2, 2, t1)
    False
    >>> contains_n(2, 1, t1)
    True
    >>> t2 = Tree(1, [Tree(2), Tree(1, [Tree(1), Tree(2)])])
    >>> contains_n(1, 3, t2)
    True
    >>> contains_n(2, 2, t2) # Not on a path
    False
    '''
    if n == 0:
        return True

    elif _____:
        return _____

    elif _____:
        return _____

    else:
        return _____
```

```

if n == 0:
    return True
elif t.is_leaf():
    return n == 1 and t.label == elem
elif t.label == elem:
    return any([contains_n(elem, n - 1, b) for b in
                t.branches])
else:
    return any([contains_n(elem, n, b) for b in
                t.branches])

```

Base cases: The simplest case we have is when $n == 0$, or when we want at least 0 instances of `elem` in `t`. In this case, we always return `True`. The other simple case we consider is when the tree is only a leaf — there is nothing left to recurse on. In that case, we simply check to see that both $n == 1$ and that `t.label == elem`, meaning that we have one element left to satisfy, and the leaf label satisfies the final element we are looking for. If we have more elements to search for (ie. $n \geq 1$), then we will not satisfy that many elements at the leaf node; conversely, if we have fewer (ie. $n == 0$), then the case would already be covered by the first base case.

Recursive cases: If the current node isn't a leaf, then there's two different cases we should consider. Either the label of the current node is equal to `elem` or the label is not equal to `elem`. For the former, we would have to search for n more elems in each branch of `t` and return `True` if any of the branches contain n elems. For the latter, we would have $(n - 1)$ elements remaining, so we would search for $(n - 1)$ more elems in each branch of `t` and return `True` if any of the branches contain $(n - 1)$ elems. Since there is not room to do a for loop, we can use a list comprehension to recursively call the function on each branch. Thus, our two list comprehension statements would be `[contains_n(elem, n, b) for b in t.branches]` and `[contains_n(elem, n - 1, b) for b in t.branches]`. To determine if any of the branches contain either n elems or $(n - 1)$ elems, we can check if there's a `True` element in the respective lists.

3. Produce the environment diagram and output that result from executing the code below.

```
def jerry(jerry):  
    def jerome(alex):  
        alex.append(jerry[1:])  
        return alex  
    return jerome
```

```
ben = ['nice', ['ice']]  
jerome = jerry(ben)  
alex = jerome(['cream'])  
ben[1].append(alex)  
ben[1][1][1] = ben  
print(ben)
```

<https://goo.gl/uhSCLR>

Print output (drag lower right corner to resize)

```
['nice', ['ice', ['cream', [...]]]]
```

